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Probability Distribution Function

```
1  def get_pdf_probability(dataset, startrange, endrange):
2  
3      from matplotlib import pyplot as plt
4      from scipy.stats import norm
5      import seaborn as sns
6  
7      # Plot histogram with KDE
8      sns.histplot(dataset, kde=True, line_kws={'color': 'blue'}, color='green')
9  
10     plt.axvline(startrange, color='red')
11     plt.axvline(endrange, color='red')
12     plt.show()
13 
14     # Calculate mean and std
15     sample_mean = dataset.mean()
16     sample_std = dataset.std()
17 
18     print(f"Mean = {sample_mean:.3f}")
19     print(f"Standard Deviation = {sample_std:.3f}")
20 
21     # Define normal distribution
22     dist = norm(sample_mean, sample_std)
23 
24     # Probability between start and end range
25     probability = dist.cdf(endrange) - dist.cdf(startrange)
26 
27     print(f"Probability between range ({startrange}, {endrange}): {probability:.4f}")
28 
29     return probability
```

Line 01: Function definition

Line 03: Import matplotlib library for plot the graph.

Line 04: Import scipy.stats library for create normal distribution.

Line 05: Import seaborn library for plot the histogram.

Line 08: Plot the histogram, by adding kde=True will draw the curve.

Line 10 & 11: Define the start and end range to check the probability and marked as red line.

Line 18 & 19: Display the mean and std deviation values.

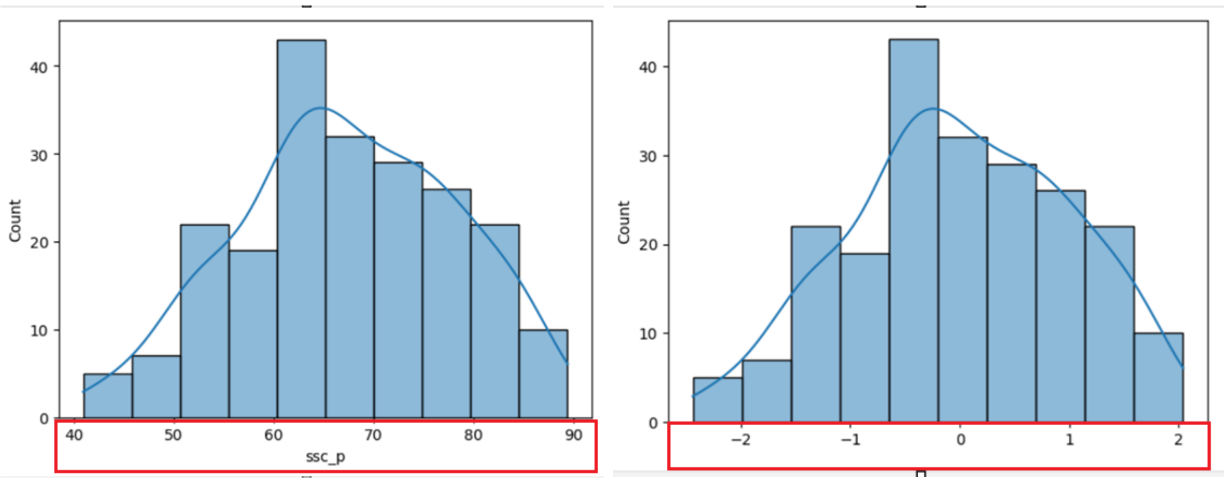
Line 22: Create the normal distribution using scipy.stats library.

Line 25: Calculate the cumulative density function for end and start range and then get the gap value by subtracting.

Line 27: Display the full information (start, end range and probability in 4 decimal places)

Line 29: Return the probability.

Standard Normal Distribution



When applying standardization to normal distribution, we can get standard normal distribution. Its range will be -3 to +3 as indicated above graphs.

```
1   def standard_normal_distribution_graph(dataset):
2       import seaborn as sns
3   
4       mean = dataset.mean()
5       std = dataset.std()
6   
7       values = [i for i in dataset]
8       z_score = [((value - mean)/std) for value in values]
9   
10      sns.histplot(z_score, kde=True)
11  
12      sum(z_score)/len(z_score)
```

Line 01: Function definition

Line 02: Import seaborn library for plot the graph.

Line 04: Get the mean value. It’s required for z-score calculations.

Line 05: Get the standard deviation. It’s required for z-score calculations.

Line 07: Assing the dataset values to list (this in called inline loop)

Line 08: Calculate the z-score value for each value in the dataset (inline loop).

$$z = \frac{x - \mu}{\sigma}$$

Line 10: Using the seaborn library plot the histogram graph and the kde attribute add a smooth, continuous probability density curve over the top of the bars.

Line 12: Calculate the final z-score